# SARAVANAN KUMAR

Senior Backend Engineer | Lead Backend Engineer | Staff Software Engineer

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## PROFESSIONAL SUMMARY

Senior Backend Engineer and technical owner with 10+ years building enterprise Java platforms at Oracle, Bosch, and Works Applications. Leads architecture and delivery for an event-driven OCI consumption platform spanning seven services and Kafka-Spark-OpenSearch pipelines processing up to 136M records. Brings hands-on strength in distributed systems, cloud data platforms, production operations, and engineering leadership.

## TECHNICAL SKILLS </>

**Backend & Architecture:** Java 8/11/17/21, Spring Boot, REST APIs, Microservices, Distributed Systems, System Design, Event-Driven Architecture, CloudEvents

**Data, Streaming & Search:** Kafka, Apache Spark, Elasticsearch, OpenSearch, Oracle Database, SQL, OCI Data Flow (managed Spark/EMR-equivalent)

**Cloud & DevOps:** Oracle Cloud Infrastructure (OCI), Docker, Kubernetes/OKE (EKS-equivalent), Helm, CI/CD, Maven, Git, Liquibase

**Observability & AI:** Grafana, Micrometer, Python, FastAPI, XGBoost, Vector Databases, GenAI

## PROFESSIONAL EXPERIENCE

### ORACLE INDIA PRIVATE LTD | Bangalore | Sep 2021 - Present

**Senior Software Engineer - IC-III | Mar 2024 - Present**

Oracle Fusion CRM - Sales Planning Application

- Lead technical design and cross-team delivery for a seven-service, event-driven OCI consumption platform supporting metrics, indexing, search, integrations, extensions, Spark processing, and notifications.
- Improved OpenSearch indexing throughput by optimizing Kafka consumer parallelism and bulk indexing, reducing a 136M-record run from about 3 hours to 2.5 hours.
- Designed resilient monthly and rolling-quarter orchestration with independent Spark extraction, controlled indexing, safe alias promotion, recovery controls, and safeguards against overlapping or out-of-order loads.
- Delivered customer-facing search, integration, and extension capabilities through REST APIs, OAuth integrations, approval workflows, spreadsheet uploads, and Kafka-driven events.
- Set technical direction for CI/CD, containerization, Kubernetes deployment, observability, production incidents, architecture reviews, code reviews, and mentoring.

**Senior Software Engineer - IC-II | Sep 2021 - Feb 2024**

Oracle Fusion CRM - Sales Planning Application

- Established the platform data-ingestion and search foundation by designing its materialized-view extraction and Elasticsearch indexing workflow.
- Built asynchronous Kafka services and helped evolve the platform from materialized-view extraction to Spark-based processing.
- Established deployment foundations by containerizing backend services and deploying them to OCI-managed Kubernetes clusters.
- Developed and maintained supporting Oracle database components, including tables, materialized views, audit and history records, scheduler jobs, reconciliation logic, and Liquibase deployments.

### BOSCH BGSW (formerly RBEI) | Coimbatore | Jan 2018 - Aug 2021

**Senior Software Engineer - Product Development**

- Built automotive firmware console tooling for HEX-file modification, CRC validation, and SHA integrity checks using Java security APIs.
- Built an automotive MDX-to-MSR conversion tool for embedded microcontroller build pipelines and streamlined source-data preparation for firmware builds.
- Received the SPOT Award for product-development performance.

### WAP (formerly IVTL) | Chennai | Jul 2016 - Dec 2017

**Software Engineer - Full Stack**

- Developed employee-management workflows in HUE, an enterprise ERP product for Japanese companies, using Java 8, Spring MVC, JavaScript, and Elasticsearch.
- Delivered employee-management functionality from business requirements through implementation and maintained Elasticsearch-backed employee search using Git-based development.

## SELECTED INNOVATION

**AI Agent for G2M Usage Load Monitoring | 2025 - Present**

MCP, ChatGPT Workspace Agents, Kubernetes/OKE (EKS-equivalent), Oracle DB, OpenSearch, Kafka, RBAC, Outlook

- Built a read-only MCP monitoring service that correlates Oracle DB, OpenSearch, Kafka, and Kubernetes health data to identify stalled, delayed, failed, or completed G2M usage loads.
- Automated daily health checks and actionable Outlook reports through a ChatGPT Workspace Agent, while keeping production credentials inside the MCP service and enforcing read-only Kubernetes RBAC.

**Extension Decision Recommendation POC | 2025 - Present**

Python, FastAPI, XGBoost, Probability Calibration, Drift Monitoring

- Contributed to a Python/FastAPI ML service using XGBoost, probability calibration, precision-based thresholds, drift monitoring, audit records, and human-review guardrails to recommend active trial plans and permitted extension options for SPM decision workflows.
- Contributed recommendation-review, cancellation, workflow-invocation, and callback APIs that preserve an auditable human-review path for extension decisions.

**GenAI Chatbot - URL and Opportunity Creation | Aug 2024 - Nov 2024**

BlenderBot, Vector Database, JavaScript, Spring Boot REST APIs

- Led a conversational-AI POC integrating BlenderBot and vector retrieval for sales-rep intent recognition; implemented JavaScript workflow UI and Search Service REST APIs for chatbot-guided URL and opportunity creation.

## EDUCATION

**B.E., Computer Science Engineering** - Sri Krishna College of Technology, Coimbatore | 2012 - 2016

## CERTIFICATIONS

Oracle Vector Database AI Search | Oracle Generative AI Professional
Oracle AI Foundations Associate | Oracle OCI Foundations Associate

## AWARDS

**STAR Performer - Oracle:** Recognized for delivering a critical feature ahead of schedule.
**SPOT Award - Bosch:** Recognized for outstanding product-development performance.